

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT (open to sponsorship)

PROFESSIONAL SUMMARY

Full-stack Applied AI engineer who writes production TypeScript and Python and translates customer requirements into shipped systems. Built and published a multi-provider browser extension (Manga Lens, Chrome Web Store) integrating four AI vision providers behind one common interface - the same enterprise-integration pattern Profound's Fortune 500 clients need. Shipped React/TypeScript/Vite full-stack apps (Dream Decoder) with structured prompt orchestration (~30% contextual alignment gain), packaged production LLM inference as FastAPI services in Docker (~30% defect reduction), and built RAG retrieval pipelines with ~35% retrieval-precision gain and >30% hallucination reduction. Master's in Computer Science (Kent State). Open to relocating to NYC/SF; US-based, F-1 OPT (open to sponsorship).

CORE COMPETENCIES

TypeScript • React • Next.js-Ready • Python & FastAPI • PostgreSQL • REST APIs & System Integration • Customer-Facing Delivery • Executive Stakeholder Communication • Production Code under Pressure • Multi-Provider AI Integration • RAG & Retrieval Engineering

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present | USA

Jr. AI/ML Engineer Trainee - Full-Stack Production AI Integration

- Built **RAG retrieval systems** with recursive semantic chunking and transformer embeddings; improved contextual retrieval precision ~**35%**, reduced irrelevant retrieval >**30%**, and exceeded **90%** response alignment in evaluation - the same retrieval/ranking discipline AEO/AI presence platforms rely on.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects ~**30%**.
- Built **agentic LLM workflows** with structured reasoning, tool discipline, and grounding rules; improved agent response stability ~**25%**.
- Designed **EHR preprocessing pipelines** (Pandas/NumPy) over appointment histories, support ticket logs, claims data; raised dataset reliability above **98%**, cut downstream model instability ~**40%**.
- Wrote **stakeholder-facing system-limitation docs** and evaluation audit trails - translating model behavior for non-technical executives.

Suvidha Foundation

Jan 2022 - Mar 2022 | India

Machine Learning Engineer - Applied NLP, Video Analytics

- Built a transformer-based summarization system over 5,000+ sessions; hierarchical long-context summarization cut manual review time **60-70%**.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~**30%**, raised answer accuracy >**85%**.
- Deployed the stack as a **Flask** API with a lightweight web interface for non-technical users.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023 | India

Database & DevOps Performance Engineer (Intern) - Enterprise ERP, Oil & Gas

- Owned **PostgreSQL / T-SQL** tuning on the Synthesis Order-to-Cash platform (contracts, nominations, allocations, invoicing); cut query execution time ~**20%** and retrieval latency ~**25%**.
- Built **CI/CD pipelines (Jenkins)** for version-controlled SQL deploys; reduced deployment errors >**30%**, improved release cycle time ~**35-40%**.
- Built **SQL Server DMV + Grafana** performance dashboards tracking long-running queries, deadlocks, and CPU/I/O contention - exec-readable BI signal for ops.
- Supported **RBAC + audit logging** for financial modules in a compliance-sensitive enterprise environment.

PROJECTS

Dream Decoder - Full-Stack FastAPI + React/TypeScript Multimodal App

Full-stack **FastAPI + React/TypeScript/Vite/Tailwind** app coordinating multimodal APIs (Perplexity Sonar for interpretation, Replicate image models for visuals). Introduced intermediate **structured prompt transformation layers** - improved contextual alignment ~**30%** and first-pass image success ~**25-30%** over naïve direct-prompt orchestration. Demonstrates building production full-stack TS/Python under one architectural roof.

FastAPI, React, TypeScript, Vite, Tailwind, structured prompt orchestration

Manga Lens - Chrome Extension Integrating 4 AI Vision Providers (shipped, Chrome Web Store)

Full-stack browser extension built independently - Manifest V3, TypeScript content scripts, service workers, **captureVisibleTab** for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v) behind one interface, with **provider-specific payload handling** (WebP cloud, JPEG local) and a seven-day translation cache - the same enterprise-integration pattern Profound's Fortune 500 clients need.

TypeScript, Manifest V3, service workers, multi-provider AI integration

Agentic Healthcare Claims & Fraud Risk Intelligence System

Multi-agent pipeline with **schema-validated JSON contracts between agents** to prevent cascading hallucinations; RAG-grounded validation against vector-indexed policy docs, ANN duplicate detection, and explainable, audit-ready reasoning traces - adjacent to Profound's data-pipeline + dashboard reporting work for executive stakeholders.

Multi-agent, JSON schema contracts, RAG, vector search, audit-ready reasoning

Patient Records Management Web App - Relational Data Modeling

Bottle + SQLite relational CRUD app with **normalized patients/diagnoses schema**, foreign-key constraints, application-layer cascade deletes, and **LEFT JOIN + GROUP_CONCAT aggregation** for single-query patient+diagnosis retrieval - the database modeling skills underpinning Postgres + time-series work.

SQL relational modeling, FK constraints, JOIN aggregation

EDUCATION

Master's Degree, Computer Science - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: TypeScript, React, Vite, Node.js, Python, FastAPI, Flask, SQL (PostgreSQL, T-SQL, SQLite)

LLM & GenAI: RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, grounding, embeddings, vector/semantic search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex

Infra & DevOps: Docker, Jenkins, CI/CD, RESTful APIs & system integration, observability/logging, Grafana

Customer / Communication: Executive stakeholder docs, system-limitation framing, founder background, multi-stakeholder coordination